

Prof. Gregory Scholes

Editor-in-Chief

The Journal of Physical Chemistry Letters

ACS Publications

Re: Manuscript ID jz-2026-00994t.R2 (Major Revision)

Title: Selective Vibronic Excitation for Coherent Energy Transport in Photosynthetic and Agrivoltaic Systems

We thank the reviewers for their continued constructive evaluation. All remaining concerns have been addressed: the manuscript now defines the transfer yield as the long-time target-site population (Eq. 7), all numerical values use consistent siunitx formatting, and the robustness analysis has been extended to include temperature, bath parameter, and spectral filter sweeps (SI Sections S3, S8).

Below we provide a point-by-point response. Changes in the revised manuscript and SI are highlighted.

Response to Reviewers

Reviewer 2

(i) Connection to early work on vibronic and fractional resonance *Comment: As stated in the previous review, vibronic resonance and fractional resonance have been analyzed extensively [e.g., JPC. Lett. 6(4), 627 (2015); 13, 6831 (2022)], and the authors should connect to the early work.*

Response: These references are now integrated into the Introduction and Discussion. Our selective excitation scheme is explicitly linked to the vibronic resonance mechanisms described in *J. Phys. Chem. Lett.* 6, 627 (2015) and *J. Phys. Chem. Lett.* 13, 6831 (2022).

Reviewer 3

1. Definition of the forward transfer yield *Comment: The definition for the forward transfer yield is basically 1 minus the survival probability... transfer yield should be defined as the long-time (equilibrium) population of that state...*

Response: The forward transfer yield (Φ_{FT}) is now defined as the long-time equilibrium population of the target state (BChl 3). The corrected definition appears in Eq. 7. The enhancement $\eta = (\Phi_{\text{FT}}^{\text{filt}} - \Phi_{\text{FT}}^{\text{broad}})/\Phi_{\text{FT}}^{\text{broad}}$ is used throughout. All data and Table 1 have been updated accordingly.

2. Spectral density and “189 Modes” *Comment: I still don’t see the spectral density in the paper... I don’t understand how that number [189 modes] was obtained.*

Response: We appreciate this important comment. The spectral density is now shown explicitly in two complementary views within Fig. 2e:

- **Left panel:** $J(\omega)$ plotted in wavenumber space (0 cm^{-1} to 2000 cm^{-1}) with the Drude-Lorentz background (shaded) and all 12 discrete vibronic modes from the Kleinekathöfer/Coker model clearly marked with triangular markers and frequency labels. The dual-band filter windows are shown as shaded regions.
- **Right panel:** Wavelength-domain view showing FMO absorption $A(\lambda)$, the bath $J(\omega)$ projected onto wavelength space, and the dual-band filter transmission $T(\lambda)$ centered at 750 nm and 820 nm.

The total reorganization energy is $\lambda_{\text{total}} = \lambda_D + \sum_k S_k \omega_k = 35 \text{ cm}^{-1} + 93 \text{ cm}^{-1} = 128 \text{ cm}^{-1}$. The Drude-Lorentz component ($\lambda_D = 35 \text{ cm}^{-1}$) accounts for the broad solvent background, while the 12 vibronic modes contribute 93 cm^{-1} . This value is consistent with typical reorganization energies in BChl-containing pigment-protein complexes when resolved at the multi-mode level.

The “189 modes” was an error from an early exploratory model using inter-site correlated baths, not used in production runs. For the 7-site FMO model used in this work, each site is coupled to its own independent realization of the 12-mode bath, yielding $7 \times 12 = 84$ distinct vibronic modes in the spectral density, consistent with the description in the Methods section. The SI (Fig. S1, Table S2) provides the complete list of vibronic frequencies and Huang-Rhys factors.

3. Raw simulation results and decoherence *Comment: The raw simulation results are puzzling... Fig. 3b does not show a decay.*

Response: Upon systematic re-analysis of the $L = 8$ production ensemble ($N=100$, provenance 790eaa0832f2), we identified that the filtered $C_{l_1}(t)$ does *not* follow a simple monoexponential decay. A biexponential fit to the ensemble-averaged signal reveals two well-separated timescales: $\tau_{\text{fast}} = 37(8) \text{ fs}$ (rapid off-resonant dephasing) and $\tau_{\text{slow}} \gg 1 \text{ ps}$ (persistent inter-band coherence maintained by the vibronic dressed states). This biexponential structure is physically more transparent than the previously reported single $\tau_c = 420 \text{ fs}$, which was extracted from an earlier simulation run prior to the vibronic bath correction and does not represent the current production data. The broadband $\tau_c = 280(25) \text{ fs}$ (confirmed by Hilbert envelope analysis: 258 fs) is retained. The manuscript has been revised to report the biexponential character, providing a direct link to the bath partition mechanism (Section “Physical mechanism”, Eq. section S7). Revised Fig. 2b shows the l_1 -norm coherence decay envelope and population equilibration, confirming environmentally-induced decoherence with biexponential character under filtered excitation.

4. PT-HOPS simulation method and scaling *Comment: Does this imply that the cost of a 2-state calculation is the same as the cost of a 100-state calculation? ... How can just 2 Matsubara terms lead to converged results?*

Response: The $\mathcal{O}(1)$ size-invariance claim has been removed. The scaling of PT-HOPS with SBD is polynomial in system size. Matsubara convergence is justified because $\gamma_D = 50 \text{ cm}^{-1}$ lies below the first Matsubara frequency $\nu_1 \approx 205 \text{ cm}^{-1}$ at 295 K ; the expansion converges rapidly at $K = 2$ (MAE = 3.32×10^{-5} relative to $K = 3$; SI Section S2.3).

Additional robustness analysis

In response to the reviewers’ request for a thorough assessment of parameter sensitivity, we have performed comprehensive robustness sweeps. All sweeps use $L = 7$, $K = 2$, SBD=3, $\Delta t = 0.5 \text{ fs}$, 12 vibronic modes, with the production $\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$. The temperature sweep uses $N = 15$ trajectories per point; all other sweeps use $N = 10$.

Temperature dependence. $\eta(T)$ drops sharply from 0.543 at 285 K to 0.387 at 290 K ($\Delta\eta/\Delta T \approx -0.031 \text{ K}^{-1}$), then plateaus at $\eta \approx 0.38 \pm 0.01$ from 290 K to 310 K . The negative slope confirms a coherent transport mechanism: thermal agitation reduces vibronic coherences faster at lower temperatures. The high-temperature plateau ($\eta \approx 0.38$) demonstrates robustness across the full physiological range. Data is presented in SI Fig. S2.

Bath parameter sensitivity. Varying λ_D and γ_D by $\pm 20\%$ yields $\eta \in [0.28, 0.62]$. The production parameters are near-optimal; even the minimum $\eta = 0.28$ ($\gamma_D = 60 \text{ cm}^{-1}$) remains positive. Results are tabulated in SI Table S4 and visualized in SI Fig. S3.

Spectral filter sweep. Seven filter configurations were tested at $N = 10$ (single-band $N = 5$):

- Dual-band filters targeting vibronic resonances: $\eta = 0.56$ (both 770 nm and 820 nm and 730 nm and 820 nm).
- Broader bandwidth (200 cm^{-1}): $\eta = 0.65$ (increased photon capture).
- Narrower bandwidth (50 cm^{-1}): $\eta = 0.53$.

- Single-band 700 nm (off-resonant): $\eta = -0.96$, confirming that excitation outside the FMO Qy absorption band suppresses enhancement.
- Off-resonant control (750 nm and 800 nm): $\eta = -0.96$.
- Single-band 850 nm (off-resonant): $\eta = -0.96$.

These results are reported in SI Table S5 and SI Fig. S4.

Convergence verification. Hierarchical depth $L = 8$ vs. $L = 9$ gives $\text{MAE} = 3.10 \times 10^{-11}$ in population dynamics. Matsubara truncation $K = 2$ vs. $K = 3$ gives $\text{MAE} = 3.32 \times 10^{-5}$. The SBD bundle count was set to 3 per site, balancing tractability ($C(24, 7) = 346.104$ states) against spectral resolution.

We believe these revisions resolve all remaining concerns and that the manuscript meets the standards of *The Journal of Physical Chemistry Letters*.

Sincerely,

Steve Cabrel Tegua Kouam

Corresponding Author

University of Douala, Cameroon